

ANALYSTLAB AFRICA
EXPERIENCE LAB INTERNSHIP PROGRAMME
HEALTHCONNECT CLINIC EXPERIENCE LAB
WEEK 7 TESTING & REFINEMENT COORDINATION PLAN
Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Field	Detail
Project:	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Programme:	AnalystLab Africa Experience Lab
Function:	Project Management
Planning Period:	Week 7
Transition From:	Week 6 Integration → Week 7 Testing & Refinement
Document Version:	1.0
Plan Owner:	Project Management
Overall Entry Position:	AMBER — CONTROLLED TESTING
Status:	Controlled Working Plan

1. PURPOSE

This plan establishes the coordination, sequencing, evidence and acceptance controls for Week 7 Testing & Refinement across the four HealthConnect workstreams:

- Data Analytics
- Data Science
- Machine Learning / ML Engineering
- Generative AI

It converts the Week 6 integration position into a controlled Week 7 execution framework.

The plan governs:

- testing entry conditions;
- test objectives and ownership;
- cross-track testing dependencies;

- sequencing of material interfaces;
- evidence and result capture;
- defect and deviation handling;
- refinement and retesting;
- acceptance and progression;
- escalation; and
- Week 7 closeout.

It does not prescribe the technical methodology of individual workstreams. Technical teams retain responsibility for determining appropriate testing methods within their areas of expertise.

2. WEEK 7 OBJECTIVE

The objective of Week 7 is to establish whether the project's developed and partially integrated outputs are sufficiently tested, refined, evidenced and controlled to support the next stage of project progression.

The intended movement is:



Week 7 is therefore not treated as another general development cycle.

A workstream may continue technical development where required, but material changes affecting an already tested output must enter the appropriate refinement and retest control.

3. CONTROL PRINCIPLES

Week 7 will operate under six control principles:

3.1 Test before acceptance

An output should not be treated as accepted merely because development is complete.

3.2 Evidence before status

Testing claims must be supported by identifiable evidence.

3.3 Actual interfaces over simulated completion

Where a downstream integration depends on a specific upstream output, the actual output should be used for integration verification wherever practicable.

3.4 Refinement requires verification

A change should not automatically be described as an improvement. Where the change affects a testable condition, its effect must be retested.

3.5 Acceptance is level-specific

Component acceptance, interface acceptance and solution readiness are separate control states.

3.6 Uncertainty remains visible

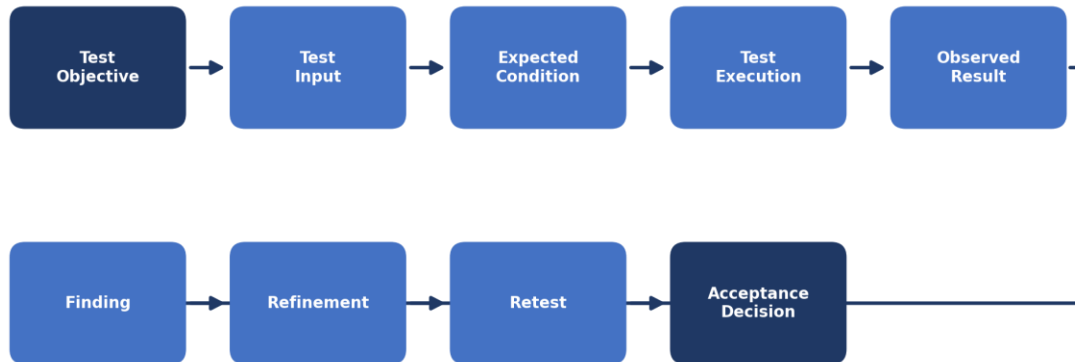
An unresolved limitation should be recorded and assessed rather than concealed by a general “ready” status.

4. WEEK 7 ENTRY POSITION

Workstream / Interface	Entry Position	Basis	Week 7 Implication
Data Analytics	Ready	Substantive analytical development completed and evidence strengthened.	Proceed to stability, interpretation and downstream-use checks.
Data Science	Conditionally Ready	Development pathway advanced; comparative execution and final candidate selection outstanding.	Testing tied to model comparison and candidate selection.
ML Engineering	Conditionally Ready	Scoring pathway and technical interface tested; actual DS candidate unavailable at Week 6 close.	Candidate integration becomes a priority dependency.
Generative AI	Conditionally Ready	Scenario evidence available; behavioural inconsistency remains open.	Targeted refinement, regression and retesting required.
Cross-Track Integration	AMBER	Material interfaces remain at different maturity levels.	Sequenced coordination required before integrated acceptance.

5. TESTING CONTROL MODEL

Each material test or refinement cycle should be traceable through the following controlled sequence:



Not every activity will require every stage. However, where a test produces a material deviation or triggers a refinement, the complete chain should be retained.

Minimum Test Record

Field	Required Information
Test ID	Unique identifier
Workstream	Responsible workstream
Interface	Downstream/upstream interface where applicable
Objective	What the test is intended to establish
Input	Data, model, output, scenario or component tested
Expected Condition	Intended behaviour / acceptance criterion
Observed Result	Actual outcome
Finding	Deviation, defect, observation or confirmation
Refinement	Change made, if applicable
Retest	Verification of the change
Evidence	Supporting artefact / result
Status	Pass / Conditional / Fail / Deferred

Field	Required Information
Owner	Responsible workstream
Readiness Effect	Consequence for downstream progression

6. TESTING STREAMS AND OWNERSHIP

Stream	Week 7 Objective	Primary Owner	PM Coordination Responsibility	Exit Evidence
Analytical Validation	Confirm stability and appropriate interpretation of priority findings.	Data Analytics	Coordinate downstream use and evidence traceability.	Validated findings, limitations and downstream-use confirmation.
Predictive Evaluation	Compare candidate models and establish selection basis.	Data Science	Track dependency, evidence sufficiency and handoff readiness.	Comparative results, selection rationale and candidate package.
Technical Integration	Integrate and verify the actual DS candidate.	ML Engineering	Coordinate DS→ML handoff and integration gate.	Actual candidate integrated and technically verified.
GenAI Behavioural Testing	Test, refine and retest defined behavioural conditions.	Generative AI	Control issue movement, retest evidence and readiness impact.	Scenario evidence, refinement record and retest results.
Cross-Track Readiness	Determine whether outputs can progress to the next stage.	PM + Workstream Owners	Consolidate evidence and determine project-level readiness.	Acceptance/readiness determination.

7. WORKSTREAM TESTING REQUIREMENTS

7.1 Data Analytics — Analytical Validation

Objective

Confirm that priority analytical findings remain sufficiently robust and appropriately bounded for downstream predictive and intervention-related use.

Week 7 Focus

- recheck material findings where necessary;
- confirm stability of priority outputs;
- distinguish priority findings from exploratory observations;
- preserve association-versus-causality boundaries;
- document subgroup limitations;
- confirm downstream analytical inputs where predictive work depends on them.

Specific Control Point

The analytical workstream should establish not simply that a finding remains statistically supported, but whether it is appropriate for the intended downstream use.

For example, statistical significance should not by itself be treated as evidence of:

- causality;
- predictive superiority;
- intervention effectiveness; or
- generalisability.

Exit Condition

Priority analytical outputs are sufficiently evidenced, appropriately interpreted and traceable into any material downstream use.

8. DATA SCIENCE — PREDICTIVE EVALUATION AND REFINEMENT

Objective

Move from prepared modelling configurations to executed comparative evidence and defensible candidate selection.

Required Week 7 Outputs

Data Science should establish, as applicable:

- candidate models evaluated;
- evaluation results;
- relevant error patterns;
- feature/input considerations;

- threshold considerations;
- model-selection basis;
- selected candidate;
- target definition;
- implementation requirements;
- known limitations.

Specific Carry-Forward Conditions

The following remain material:

1. comparative model execution;
2. final candidate selection;
3. treatment of Cancelled records under the binary target; and
4. transfer of the selected candidate to ML.

Control Boundary

The project should not infer:

development → improvement

or:

best metric → best project model

without evidence supporting the selection basis.

Exit Condition

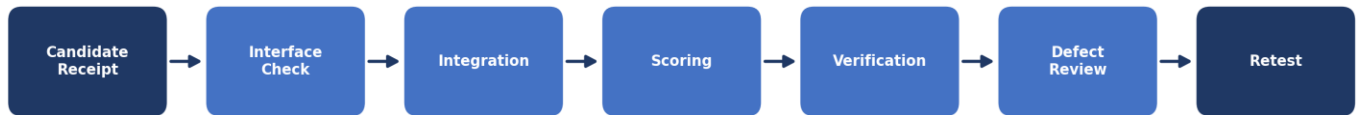
A defined candidate has been evaluated, selected and documented sufficiently for the ML implementation handoff.

9. ML ENGINEERING — TECHNICAL INTEGRATION AND VERIFICATION

Objective

Verify the actual selected Data Science candidate within the ML implementation pathway.

Week 7 Sequence



Required Evidence

Where applicable:

- candidate received;
- interface/schema compatibility confirmed;
- integration executed;
- scoring output generated;
- output integrity checked;
- defects recorded;
- corrections applied;
- retest completed.

Critical Boundary

The Random Forest stand-in used during Week 6 demonstrated the technical integration pathway.

It does not constitute completion of the intended DS→ML handoff.

Week 7 should therefore distinguish clearly between:

“The integration mechanism has been tested.”

and:

“The selected project model has been integrated and verified.”

Operational Readiness

Technical scoring capability remains separate from operational use.

Before risk scores are treated as an operational intervention input, the project must establish their intended interpretation and decision context.

Exit Condition

The selected DS candidate is technically integrated, verification evidence is available and material interface defects are either resolved or formally controlled.

10. GENERATIVE AI — BEHAVIOURAL TESTING AND REFINEMENT

Objective

Establish whether the GenAI capability demonstrates sufficiently consistent behaviour across the relevant safety, support and service-information scenarios to progress toward broader validation.

Priority Test Areas

Week 7 testing should include relevant scenarios covering:

- emergency escalation;
- disclaimer behaviour;
- diagnosis-related refusal;
- treatment and medication boundaries;
- pricing/service-information boundaries;
- unsupported or fabricated information;
- ambiguous health-related requests; and
- regression of previously tested behaviours.

Known Week 6 Carry-Forward

The inconsistent emergency-disclaimer behaviour remains an open refinement item.

The Week 7 objective is therefore not merely to modify the behaviour, but to determine through retesting whether the expected behaviour is consistently reproduced.

Evidence Requirement

Each material scenario should retain:



Evidence Boundary

Selected successful scenarios do not establish universal reliability.

Likewise:

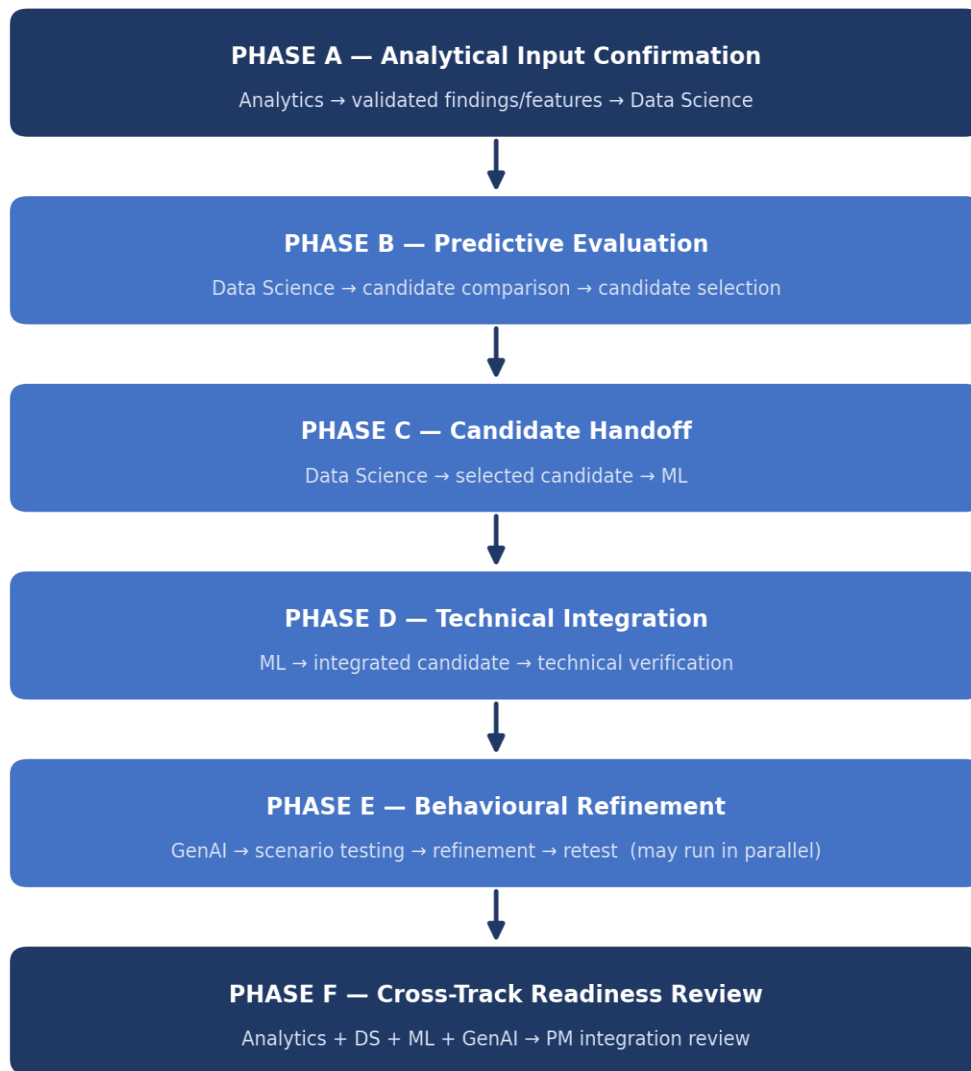
- no observed privacy failure $\neq$ privacy validation;
- scenario success $\neq$ broad behavioural validation;
- model/configuration change $\neq$ performance improvement.

Exit Condition

Material known behavioural conditions have been retested, residual limitations are documented and the evidence supports the applicable readiness decision.

11. CROSS-TRACK TESTING SEQUENCE

Testing should follow the material dependency chain rather than treating all workstreams as independent.



Phase E may proceed in parallel where it does not depend on the predictive integration sequence.

12. FINDING AND DEVIATION CONTROL

Week 7 findings should be classified according to their effect on the project.

Classification	Definition	Control Response
Observation	Noted result with no material failure against the expected condition.	Record where useful; monitor if relevant.
Deviation	Observed result differs from the expected condition.	Assess significance and determine refinement requirement.

Classification	Definition	Control Response
Defect	Material failure against a defined technical or functional condition.	Correct, retest and record closure evidence.
Integration Failure	Interface prevents the required downstream use or transfer.	Escalate through integration controls.
Acceptance Gap	Evidence is insufficient to establish the applicable acceptance condition.	Obtain additional evidence or formally defer acceptance.
Blocker	Condition prevents the planned testing or downstream activity from proceeding.	Immediate escalation and resolution.

Material issues should be maintained in the Integration Issue Log rather than duplicated in this plan.

13. REFINEMENT CONTROL

A refinement is any controlled change made in response to a test finding, defect, evidence gap or identified performance/behavioural limitation.

For material refinements, PM should retain the chain:



Refinement Status

Status	Meaning
Identified	Refinement requirement recognised.
In Progress	Change being implemented.
Implemented — Unverified	Change completed but relevant retest not yet completed.
Verified	Retest demonstrates the intended effect.
Rejected / Ineffective	Retest does not establish the intended effect.
Deferred	Refinement deliberately carried forward with documented rationale.

Status	Meaning
Closed	Evidence supports completion and acceptance.

A refinement should not be described as an improvement solely because a change was made.

14. RETEST CONTROL

Retesting is required where a material change could affect an established test condition.

Triggers include:

- correction of a material defect;
- model or configuration change;
- interface modification;
- failed acceptance condition;
- material change to test input;
- regression concern; or
- evidence that the original test is no longer representative.

Retest Requirement

The retest should establish whether:

1. the original deviation is resolved;
2. the intended behaviour is now achieved;
3. the change introduced a new adverse effect;
4. previously successful behaviour remains intact; and
5. the acceptance condition can now be considered satisfied.

15. ACCEPTANCE FRAMEWORK

Acceptance is assessed at three distinct levels.

Level	Question	Example
Component Acceptance	Does the individual output satisfy its defined condition?	DS candidate meets agreed evaluation/documentation requirements.

Level	Question	Example
Interface Acceptance	Can the receiving workstream use the output as intended?	ML receives and successfully incorporates the selected DS candidate.
Integrated Acceptance	Does the combined workflow behave as required?	Predictive output can operate within the intended downstream workflow.

Control Rule

Component acceptance does not automatically constitute integrated acceptance.

Similarly, technical integration does not automatically constitute operational readiness.

16. WEEK 7 ACCEPTANCE CRITERIA

The project should assess material outputs across five dimensions:

Dimension	Acceptance Question
Functional	Does the output perform its intended function?
Technical	Does it operate within the defined technical conditions?
Evidence	Is the evidence sufficient to support the claim being made?
Integration	Has the required downstream interface been successfully demonstrated?
Operational	Can the output be meaningfully used within the intended workflow?

The applicable dimensions will depend on the output being assessed.

17. TESTING DEPENDENCY CONTROL

Dependency	Provider	Receiver	Week 7 Requirement	Consequence if Unavailable
Validated analytical inputs	Analytics	DS	Confirm usable and traceable inputs.	Predictive development may lack confirmed analytical basis.
Selected candidate	DS	ML	Transfer actual candidate package.	Candidate integration cannot close.

Dependency	Provider	Receiver	Week 7 Requirement	Consequence if Unavailable
Technical implementation	ML	Integrated Workflow	Verify actual candidate.	Integrated technical testing delayed.
GenAI refinement evidence	GenAI	PM / Validation	Complete targeted retest.	Behavioural readiness remains conditional.
Predictive target definition	DS / PM	DS / ML	Resolve Cancelled treatment.	Target consistency remains exposed.
Acceptance conditions	PM / Workstreams	Integrated Delivery	Confirm applicable conditions before closure.	Test results may not support formal acceptance.

18. ESCALATION CONTROL

An unresolved testing condition should be escalated where it:

- prevents a downstream workstream from proceeding;
- affects a material acceptance condition;
- creates a safety or governance concern;
- repeatedly fails after refinement;
- threatens the Week 7 testing sequence;
- produces a material conflict in interpretation or ownership; or
- cannot be resolved within the responsible workstream's authority.

Escalation Principle

Escalate on material consequence, not simply on technical difficulty.

Project Management coordinates escalation and maintains visibility; technical ownership remains with the relevant workstream.

19. WEEK 7 CONTROL GATES

Gate	Control Point	Minimum Evidence	Decision
G1 — Test Readiness	Inputs and conditions ready.	Objective, input, expected condition and owner defined.	Authorise testing.
G2 — Predictive Candidate	DS candidate established.	Comparative evidence and selection rationale.	Authorise candidate handoff.

Gate	Control Point	Minimum Evidence	Decision
G3 — Technical Integration	Actual candidate integrated.	Integration and verification evidence.	Progress to integrated testing.
G4 — Refinement Verification	Material findings addressed.	Retest evidence.	Accept, defer or escalate.
G5 — Interface Acceptance	Cross-track outputs usable downstream.	Receiver acceptance and resulting-change evidence.	Confirm interface closure.
G6 — Readiness Review	Week 7 evidence consolidated.	Open findings, limitations, acceptance status and dependencies.	Determine next-stage readiness.

A gate may be conditional where controlled progression is justified but full closure has not yet been demonstrated.

20. ROLES AND ACCOUNTABILITY

Function	Accountability
Project Management	Owns coordination framework, sequencing, dependency visibility, evidence control, escalation and project-level readiness assessment.
Data Analytics	Owns analytical validation and evidence supporting analytical outputs.
Data Science	Owns predictive evaluation, candidate selection and model handoff package.
ML Engineering	Owns technical integration, scoring verification and implementation evidence.
Generative AI	Owns behavioural testing, refinement and retesting within the GenAI workstream.
Project Leadership	Owns decisions or escalations beyond workstream authority.

PM does not determine technical pass/fail outcomes on behalf of the technical owners. PM determines whether the evidence and coordination conditions required for project progression have been satisfied.

21. REPORTING AND CONTROL CADENCE

Checkpoint	PM Activity	Primary Output
Testing Start	Confirm entry conditions and active dependencies.	Test-readiness position.

Checkpoint	PM Activity	Primary Output
Integration Checkpoint	Review handoffs, test progress and material findings.	Updated coordination position.
Refinement Review	Review material changes and retest requirements.	Refinement / retest status.
Pre-Acceptance Review	Confirm evidence completeness and unresolved conditions.	Acceptance readiness position.
Week 7 Closeout	Consolidate results and carry-forward requirements.	Week 7 readiness determination.

Material blockers or safety/governance concerns should be escalated immediately rather than waiting for the next scheduled checkpoint.

22. WEEK 7 CONTROLLED DELIVERABLES

The Week 7 execution cycle is expected to produce, where applicable:

1. Data Analytics validation evidence;
2. Data Science comparative model evidence;
3. final candidate-selection record;
4. selected-model handoff package;
5. ML candidate-integration evidence;
6. scoring/interface verification evidence;
7. GenAI scenario and regression evidence;
8. refinement and retest records;
9. updated limitations;
10. acceptance determinations;
11. updated interface/dependency positions;
12. material issue updates; and
13. consolidated Week 7 readiness assessment.

These outputs should feed the existing PM control records rather than create parallel undocumented status streams.

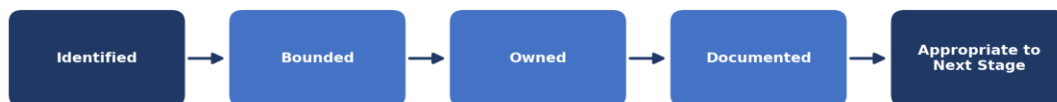
23. SUCCESS CONDITIONS

Week 7 will be considered effectively controlled when the project can demonstrate that:

- material outputs were tested against defined conditions;
- testing evidence is traceable;
- significant findings were assessed for project impact;
- material refinements were retested;
- the intended DS candidate was transferred and technically verified where applicable;
- downstream use of material analytical inputs is traceable;
- GenAI carry-forward behaviour was retested;
- unresolved conditions have explicit ownership and disposition;
- acceptance status is distinguishable from development status; and
- the project can make a defensible next-stage readiness determination.

The objective is not to eliminate every uncertainty.

The objective is to ensure that remaining uncertainty is:



24. WEEK 7 ENTRY DETERMINATION

AMBER — CONTROLLED TESTING MAY PROCEED

The Week 6 evidence supports entry into Week 7 Testing & Refinement.

Progression should remain subject to active control of:

- the DS→ML candidate-model handoff;
- predictive candidate selection;
- Analytics→DS downstream traceability;
- ML operational interpretation;
- GenAI disclaimer refinement and retesting; and
- predictive target-definition closure.

These conditions do not collectively establish a project-wide blocker at Week 7 entry.

They represent controlled integration conditions that must be managed through the testing cycle.

25. GOVERNANCE INTERFACE

This plan is the forward-looking execution control for Week 7 and operates alongside the existing PM records.

Existing Record	Relationship to This Plan
Cross-Track Coordination Record	Provides evidence of actual coordination and handoffs.
Integration & Dependency Register	Defines and monitors material dependencies.
Integration Issue Log	Controls material integration issues and their closure.
Decision Log	Records formal project decisions affecting testing or progression.
Risk & Dependency Register	Controls broader project exposure.
Project Status Update	Reports overall project health and management position.

This plan does not reproduce those records. It establishes how Week 7 testing and refinement will be coordinated against them.

26. RECORD MAINTENANCE

Plan Owner: Project Management

Technical Evidence Owners: Relevant Workstream Leads

Update Trigger: Material change to testing sequence, dependency, acceptance condition, refinement status or readiness

Review Frequency: Testing checkpoints and Week 7 closeout

Escalation Authority: Project Leadership where resolution exceeds workstream authority

The plan should be updated where a material change alters the Week 7 testing sequence or progression conditions.

27. VERSION CONTROL

Version	Stage	Change
1.0	Week 6 → Week 7	Initial controlled Testing & Refinement Coordination Plan established from Week 6 integration evidence and readiness position.

Version	Stage	Change
1.1	Week 7	Updated for actual testing, findings, refinements, retesting and changed dependencies.
2.0	Week 7 Closeout	Re-baselined following testing, acceptance and readiness review.